
Otto-von-Guericke University Magdeburg



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INF

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Simulation Project

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Abstract

This project report contains a detailed analysis of the intersection Robert-Koch-Straße, Draisweg, An der Steinkuhle, Rennetal and various experiments conducted to improve the efficiency of the existing system with some assumptions.

A conceptual model of the intersection was used as a basis to create a simulation model in AnyLogic. The model takes into account important parameters such as the traffic volume, lane configurations, traffic signal settings, vehicle speed and other real world data. Traffic surveys and data collected was used for calibration, data analysis and validation of the model.

We have used AnyLogic simulation software as a tool to test out design changes and analyze the effects of the experiment on the efficiency of the intersection. The objective of the project is to test the efficiency of the proposed experiments and provide a informed recommendation to the city of Magdeburg if it resulted in a clear improvement compared to the current state.

Output results from the experiments such as the throughput and time spent in the model were used as performance indicators. Trends were observed from these indicators and were used to give an informed decision on whether the proposed experiments can be recommended.

Zusammenfassung

Diese Untersuchung fand im Rahmen der Veranstaltung “Simulation Project” der Fakultät für Informatik der Otto-von Guericke-Universität Magdeburg statt. Hierbei wurde die in Abbildung 1 dargestellte Kreuzung analysiert, welche sich aus den Straßen “Robert-Koch-Straße”, “Rennetal”, “An der Steinkuhle” und “Draisweg” zusammensetzt. In vier Experimenten wurden die Auswirkungen von Änderungen an dem Aufbau der Kreuzung auf die Zeit, die in der Kreuzung verbracht wurde, und die Anzahl der Autos, die diese erfolgreich durchquert haben (“Throughput”), betrachtet. Eine anschließende Auswertung gibt eine Empfehlung an die Stadt Magdeburg bezüglich der weiteren Planung für die untersuchte Kreuzung.



Figure 1: Intersection overview

Als erster Schritt wurde ein Zeitplan für das Projekt entwickelt. Dabei wurden insbesondere die zu erfüllenden Arbeitspakete für bestimmte Meilensteinen analysiert und geplant. Außerdem wurde eine Kostenschätzung für die verschiedenen Meilensteine aufgestellt und eine Übersicht über den zu erwartenden Projektfortschritt erstellt.

Als Grundlage für die Simulation wurde zunächst ein Konzeptuelles Modell in Form eines Petri-Netzes entwickelt. Mithilfe des Modells konnte ein gutes Verständnis der zu untersuchenden Kreuzung entwickelt werden und es wurde ein Überblick über die benötigten Daten für eine weitere Untersuchung gewonnen. Für die Modellierung wurden einige grundlegende Annahmen getroffen. So wird erwartet, dass sich alle Verkehrsteilnehmer an die Verkehrsregeln halten und sich nur im Rahmen der existierenden Straßen bewegen. Unter anderem wurde auch das Überholen von Autos der Einfachheit halber weggelassen.

Um eine Datengrundlage für die Simulation und die Experimente zu haben, wurde an mehreren Tagen morgens von 7:00 bis 8:00 an der zu untersuchenden Kreuzung die Verkehrssituation analysiert. Dabei wurde für jede Straße ermittelt, mit welcher Wahrscheinlichkeit in die jeweils anderen Straßen abgebogen wurde. Außerdem wurde gemessen, wie viel Zeit zwischen den Ankünften der PKW in jeder Straße liegt. Probleme gab es hierbei bei der geringen Datenlage insbesondere für die Straßen “An der Steinkuhle” und “Rennetal”, da diese nur einen sehr geringen Verkehrsfluss aufweisen. Weiterhin muss beachtet werden, dass die üblichen Fehler durch die

manuelle Erfassung der Daten vorliegen. Durch die gesammelten Daten konnte dann mittels der Verwendung von statistischen Methoden wie Quantil-Quantil-Diagrammen und Chi-Quadrat-Tests die Verteilungen für die Ankunft der PKW erfolgreich bestimmt werden.

Zunächst wurde der Ist-Zustand der Kreuzung mit dem Tool “AnyLogic” simuliert. Auf Grundlage der Daten wurden die Geschwindigkeit der Autos sowie die Wahrscheinlichkeit für das Abbiegen in bestimmte Straßen als Teil des Simulationsprogramms aufgenommen. Zur Verifikation des Programms wurden diverse Maßnahmen getroffen, unter anderem wurden zunächst grundlegende Komponenten implementiert und ihre Funktionalität überprüft, bevor diese zum Simulationsprogramm kombiniert wurden (Unit Testing).

Um die Korrektheit des Simulationsmodells zu überprüfen, wurden mehrere Validierungsmethoden verwendet. In einem ersten Schritt wurde mittels einer Sensitivitätsanalyse die Augenscheinvalidität (Engl.: face validity) erfolgreich gezeigt. Anschließend wurde die Simulation mit Hilfe von Input-Output-Transformationen mit 25 Replikationen und einem Konfidenzlevel von 98% validiert.

Im Rahmen des Projektes wurden vier mögliche Anpassungen der Kreuzung analysiert. In Experiment 1 wurde untersucht, welchen Einfluss das Zusammenführen aller Straßen in einem großen Kreisverkehr hat. Kreisverkehre lassen sich aufgrund der vereinfachten Verkehrsführung für alle Verkehrsteilnehmer besser überblicken und führen somit zu einer höheren Sicherheit im Straßenverkehr. Die Ergebnisse der Simulation zeigen keine statistisch signifikante Verbesserung des Throughput im Vergleich zur aktuellen Verkehrsführung. Im Hinblick auf die in der Kreuzung verbrachte Zeit lässt sich eine leichte, praktisch nicht signifikante Verschlechterung feststellen. Auch bei einer Verdoppelung des simulierten Verkehrsaufkommens kommt es zu diesen Resultaten.

Da es sich um eine in Nord-Süd-Richtung sehr langgezogene Kreuzung handelt, wurde in Experiment 2 untersucht, welche Auswirkung das Aufteilen der Kreuzung auf zwei kleine Kreisverkehre hat. Dafür wurden im Norden der Kreuzung die Straßen “Robert-Koch-Straße”, “Rennetal” und “An der Steinkuhle” und im Süden die Straßen “Draisweg” und “Robert-Koch-Straße” über einen Kreisverkehr zusammengeführt. Auch hier führt die Veränderung der Straßenführung zu keiner statistisch signifikanten Verbesserung des Throughputs und die verbrachte Zeit erhöht sich leicht um eine praktisch nicht signifikante Differenz. Bei einer Verdoppelung des Verkehrsaufkommens verbringen die PKW signifikant mehr Zeit im System.

In Experiment 3 wurden im Norden und Süden der Robert-Koch-Straße Ampeln hinzugefügt. Da der Großteil des Verkehrs über die Robert-Koch-Straße verläuft, würde das Hinzufügen von zwei Ampeln das Überqueren der Straße für Fußgänger und Radfahrer vereinfachen. Bei der Auswertung dieses Experiments lässt sich jedoch feststellen, dass die Zeit, die die PKW im System verbringen, um mehr als 100% ansteigt.

Um den Hauptverkehrsfluss auf der Robert-Koch-Straße aufzuteilen, wurde in Experiment 4 untersucht, welchen Einfluss die Aufhebung der Einbahnstraßenregelung für den Draisweg hat. Es lässt sich sowohl für den Throughput als auch für die verbrachte Zeit keine praktisch signifikante Veränderung feststellen. Somit stellt diese Variation keine Verbesserung dar, sondern bringt nur die Nachteile einer komplexeren Verkehrssituation mit sich.

Insgesamt bewegen sich die Ergebnisse der Experimente alle im Rahmen einer deutlichen Verschlechterung bis hin zu einer maximal leichten, praktisch nicht signifikanten Verbesserung. Somit lässt sich feststellen, dass es für die aktuelle Verkehrssituation der Kreuzung keinen Mehrwert bietet, in eine Umsetzung der im Kontext dieses Projektes untersuchten Änderungen zu investieren. Auch falls es zu einem erhöhten Verkehrsaufkommen im Rahmen von Bauarbeiten

oder einer zukünftigen Veränderung der Verkehrssituation kommt, stellt der aktuelle Aufbau der Kreuzung die beste der untersuchten Alternativen dar.

Das Team konnte somit im Rahmen des Projektes ein valides Simulationsmodell der aktuellen Verkehrssituation implementieren. Nach der Untersuchung von vier möglichen Alternativszenarien konnte eine Empfehlung an die Stadt Magdeburg abgegeben werden. Der Zeitplan, das Budget und die weiteren Anforderungen wurden eingehalten.

Contents

Abstract

Zusammenfassung

List of Figures

List of Tables

List of Abbreviations

1 Introduction

1.1 Intersection description	1
1.2 Special features about the node	2
1.3 Utilized tools and platforms	2

2 Project Plan

2.1 Project phases	3
2.2 Project schedule	3
2.3 Organization of work packets	6
2.4 Cost estimation	6

3 Conceptual Model

3.1 Stochastic Petri Nets	7
3.2 Model's underlying assumptions and simplifications	8
3.3 Conceptual model using SPN	8
3.4 Quantities to be measured	10
3.5 Quantities that will be used as simulation results	10

4 Data Analysis

4.1 Data received	11
4.2 Input and Output data	12
4.3 Results of Histograms, Q-Q plots, Chi-square tests	12
4.4 Difficulties encountered while obtaining the data	15
4.5 Limitations on the accuracy or validity of the data	15

5 Simulation Program

5.1 Program concept and structure	17
5.2 Modularization	18
5.3 Differences to the conceptual model	19
5.4 Steps taken to verify the program	19
5.5 Difficulties encountered	19

6 Validation	
6.1 Quantities of interest	20
6.2 Sensitivity Analysis	20
6.3 Model parameters for sensitivity analysis	21
6.4 Sensitivity analysis results	21
6.5 Input-Output transformations and results	23
6.6 Justifications	25
6.7 Statement of confidence	25
6.8 Changes made to the model	25
6.9 Limitations to the scope of validity	25
7 Simulation Experiments	
7.1 Experiment 1 - One Large Roundabout	27
7.1.1 Idea	27
7.1.2 Results obtained after keeping the same traffic volume	28
7.1.3 Results obtained after doubling the traffic volume	29
7.1.4 Recommendations	29
7.2 Experiment 2 - Two Small Roundabouts	30
7.2.1 Idea	30
7.2.2 Results obtained after keeping the same traffic volume	30
7.2.3 Results obtained after doubling the traffic volume	31
7.2.4 Recommendations	32
7.3 Experiment 3 - Two Traffic Lights	33
7.3.1 Idea	33
7.3.2 Results and Observations	33
7.3.3 Recommendations	34
7.4 Experiment 4 - Removing One-Way Restriction	34
7.4.1 Idea	34
7.4.2 Results and Observations	35
7.4.3 Recommendations	36
7.5 Recommendation Summary	36
8 Challenges Encountered	
8.1 Challenges encountered	38
9 Project Cost Diagram	
10 Project Progress Diagram	
11 Conclusion	
A Bibliography	

List of Figures

1	Intersection overview	ii
1.1	Images of intersection with their abbreviations	1
2.1	Gantt Chart for milestone 1-4	4
2.2	Gantt Chart for milestone 5-8	5
2.3	Trello board for assignment of tasks	6
3.1	Petri net model for the intersection	9
4.1	RK-N Plots	12
4.2	RK-S Plots	13
4.3	Rennetal Plots	14
4.4	An der Steinkuhle Plots	15
5.1	Simulation model	17
5.2	Car sources	18
5.3	Bus sources	18
6.1	Throughput of the system on varying traffic volume	22
6.2	Mean time of vehicles in the system	22
6.3	Mean times for vehicles of the streets	23
6.4	Boxplot of the throughput of the systems	24
6.5	Boxplot of the time spent in the systems	24
7.1	One roundabout combining all streets	27
7.2	Confidence interval for experiment 1	28
7.3	CI for Experiment 1 with doubled traffic volume	29
7.4	Two roundabouts	30
7.5	Confidence interval for experiment 2	31
7.6	CI for Experiment 2 with doubled traffic volume	32
7.7	Two traffic lights on Robert-Koch-Straße	33
7.8	CI for Experiment 3	34
7.9	Removing one way restriction	35
7.10	CI for Experiment 4	36
9.1	Project cost diagram	39
10.1	Project progress diagram	40

List of Tables

2.1	Milestone leads with milestone cost estimation	6
4.1	Chi-square test values for RK-N	13
4.2	Chi-square test values for for RK-S	13
4.3	Chi-square test values for Rennetal	14
4.4	Chi-square test values for An der Steinkuhle	15

List of Acronyms

Acronym	Meaning
CI	Confidence Interval
D	Draisweg
R	Rennetal
RK-N	Robert-Koch-Straße North
RK-S	Robert-Koch-Straße South
S	An der Steinkuhle
SPN	Stochastic Petri Nets

Introduction

The goal of this simulation project is to boost traffic flow while decreasing the possibility of accidents at the intersection of Robert-Koch-Straße, Rennetal, An der Steinkuhle and Draisweg.

This report and the work done follows the theoretical concepts and applications mentioned in the Introduction to Simulation course. [1]

1.1 Intersection description

The abbreviations RK-N, R, D, RK-S, and S are used to identify the five roads at the intersection of Robert-Koch-Straße (North - entering the system from top), Rennetal, Draisweg, Robert-Koch-Straße (South - bottom-up entry into the system), and An der Steinkuhle. The illustration 1.1a below shows these as well .



(a) Intersection with abbreviations



(b) Picture of real world intersection

Figure 1.1: Images of intersection with their abbreviations

1.2 Special features about the node

Every street ought to have a 30 km/h speed restriction for cars. Robert-Koch-Straße sees the majority of the traffic. The number of pedestrians and bicycles varies according to the season. Although, Draisweg is a one-way street and no vehicles enters our intersection from Draisweg, a few bicycles and pedestrians travel from there and enter our intersection. Since, there is no zebra crossings or traffic signal in our intersection, the pedestrians and the bicyclists cross the streets, when they feel secure to do so. There is only one bicycle track which goes on the Robert-Koch-Straße. Other streets do not have any special bicycle tracks. Another special feature, is that the traffic symbols are hidden and not seen clearly. There is also a blind spot for vehicles turning from the Robert-Koch-Straße to An der Steinkuhle.

1.3 Utilized tools and platforms

We used a variety of platforms for collaboration on our project.

- To provide an overview of each milestone, take notes, and engage in brainstorming, **Miro board** was employed.
- **Trello** was used to manage project activities, allocate tasks, and monitor progress.
- **Clockify** was used to keep track of the amount of time spent on each task and calculate the cost.
- **Google drive** was used to share files.
- The project report and project presentations were created using **Overleaf**.
- **Whatsapp** was used for faster communication.

2

Project Plan

In this milestone, we planned our project schedule, deadline and estimated the cost incurred for each task. We used various collaboration tools in order to organize the tasks. We divided the major milestones of this project, into smaller tasks called work packets. These work packets will be shown and described in the Gantt chart later.

2.1 Project phases

The project is divided into several phases depending on each major milestones. Each phase has a designated milestone leader who is responsible for assigning tasks, establishing deadlines, and keeping an eye on the budget. The following are the phases along with their respective leaders:

- **Project Leader** - Sarah Köneke
- **Conceptual Model** - Shruti Kulkarni
- **Data Acquisition** - Dhanashree Gunda
- **Chief Software Architect** - Vanessa Tojan
- **Validation and Quality Control** - Nithin Baiju
- **Experimental Design** - Yashas Manjunath

2.2 Project schedule

Each team member is in charge of a certain milestone for which they are accountable and answerable. The respective milestone leaders had to divide the tasks into small and manageable work packets, who then assigned each work packet to a specific team member. The figure 2.1 and 2.2, shows a Gantt chart for the project schedule. We depicted the milestone work packets, the tasks assigned to each team member, and the deadlines for each work packet in it, which should not exceed more than 7 days.

Simulation Project

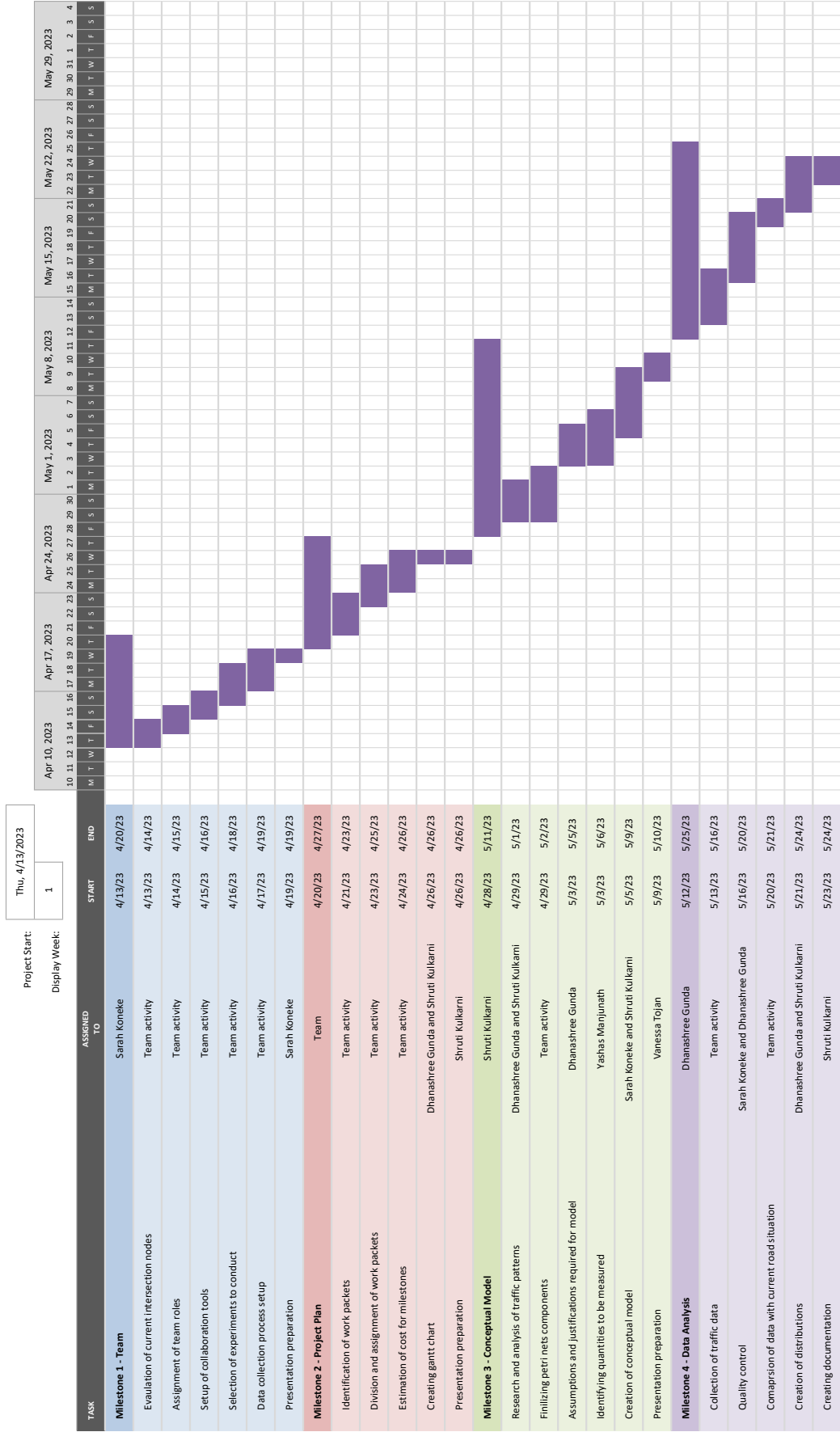


Figure 2.1: Gantt Chart for milestone 1-4

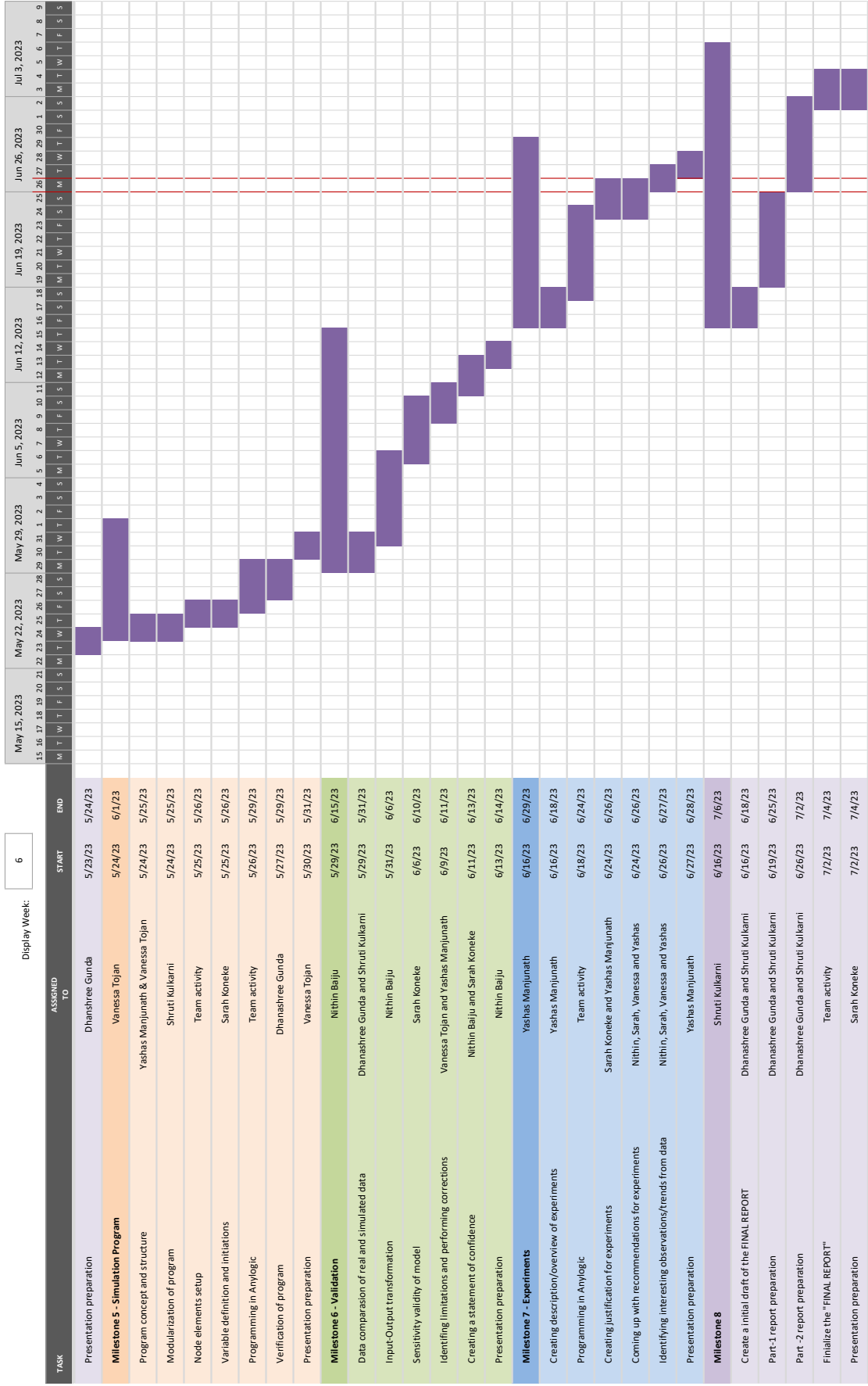


Figure 2.2: Gantt Chart for milestone 5-8

2.3 Organization of work packets

Using Trello board as the visual tool for collaboration, we established a project timeline along with the tasks in this milestone. We used the Trello board to plan, organize, schedule, and assess the tasks assigned to team members who worked independently.

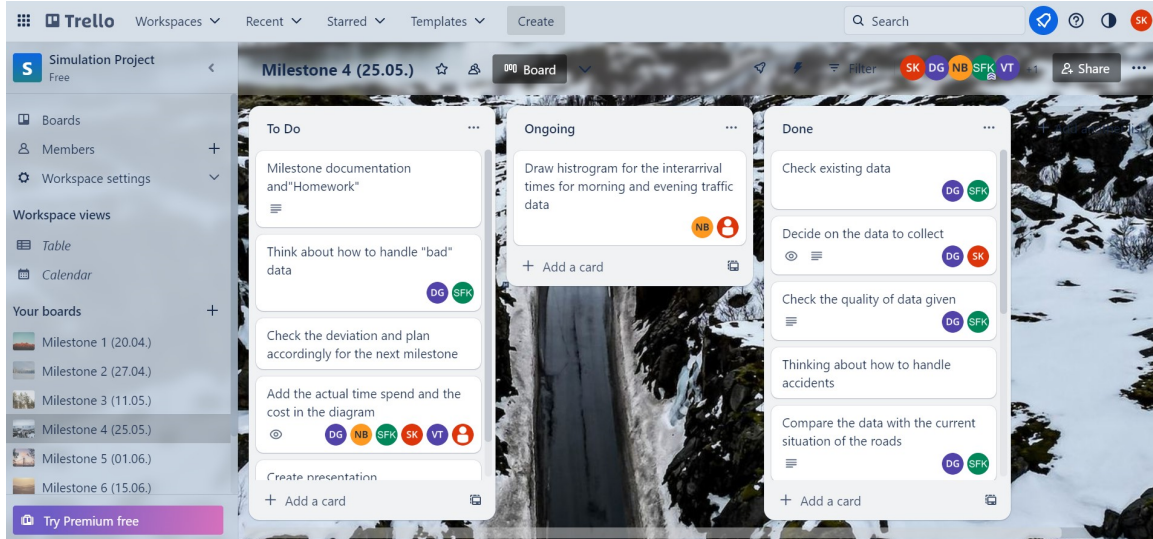


Figure 2.3: Trello board for assignment of tasks

2.4 Cost estimation

We estimated the project on a milestone-by-milestone basis. The cost estimates are summarized in the table number.

Milestone	Milestone Lead	Expected Cost (€)
1	Sarah Köneke	2.700
2	Shruti Kulkarni	3.000
3	Shruti Kulkarni	7.200
4	Dhanashree Gunda	9.000
5	Vanessa Tojan	10.200
6	Nithin Baiju	12.600
7	Yashas Manjunath	7.200
8	Dhanashree Gunda	4.800

Table 2.1: Milestone leads with milestone cost estimation

The budget of 60.000€ was allotted to our team, and the estimated cost was 56.600€, leaving 3.300€ in the budget.

3

Conceptual Model

A conceptual model is a graphical representation of the real system. It helps us gain a basic understanding of the system and can be used as a basis for discussions to solve the real life scenarios.

3.1 Stochastic Petri Nets

We used a Stochastic Petri net as a conceptual model in this simulation project. A SPN consists of the following -

- **Tokens/Agents** - representing objects (entities). They are drawn as dots located in places.
- **Places** - representing locations or states. They are drawn as circles.
- **Transitions** - represent activities or events. They are classified into two types -
 - **Immediate Transitions** - representing events (without activities) which fire immediately when they are enabled. They are drawn as solid rectangles.
 - **Timed Transitions** - representing activities and the events which end them. They are drawn as open rectangles.
- **Arcs** - are connectors between places and transitions which give the direction to the flow of the model.
 - **Input Arcs** - join places to transitions which determine enabling of transitions.
 - **Output Arcs** - join transitions to places which determine number tokens created by firing.
 - **Inhibitor Arcs** - join places to transitions which disable transitions when enough tokens are in place.
- **Events** - are the occurrences which changes the state of the system. Events are classified as primary and secondary.

- **Primary Event** - is an event whose occurrence is scheduled at a certain time.
- **Secondary Event** - is an event which is triggered by a certain condition becoming true.

3.2 Model's underlying assumptions and simplifications

Before designing and building the model, some assumptions were made to simplify the model design. Our first assumption was that every vehicle on the street, in our intersection will comply with the traffic laws and signals. We also assumed that all the vehicles, regardless of it being a car, bus or truck, have the same speed throughout the intersection and are represented as agents. Due to the streets being small, we assumed the vehicles do not overtake each other in the traffic. Instead, they form a queue in case of traffic. In our model, for simplification, we have considered cyclists as vehicles as well.

3.3 Conceptual model using SPN

This Petri net model was designed on the miro board platform. The Petri net of the entire system is shown in the figure 3.1. When an agent enters the system, it goes in the pool/queue of a road. The agent either goes left or right and moves to the respective place in the model. The yield is given to all the vehicles in R-K-Straße. All other agents from Rennetal and An der Steinkuhle have to wait, look at the street for any agent on R-K-Straße and then proceed with caution. Once the agent enters the intersection, it exits the system within a certain time frame. There is also a bicycle lane, which is represented by a separate Petri net parallel to and overlapping the R-K-Straße. Since Draisweg is a one-way street, the agents can only exit the system through it.

Each agent enters the system and is placed in the pool/queue through a timed transition with an inter-arrival time, the data of which was collected as explained in section 4. It is immediately moved to the intersection according to the system of queue. It then stays in the system for a certain time until the time transition is activated and eventually exits the system.

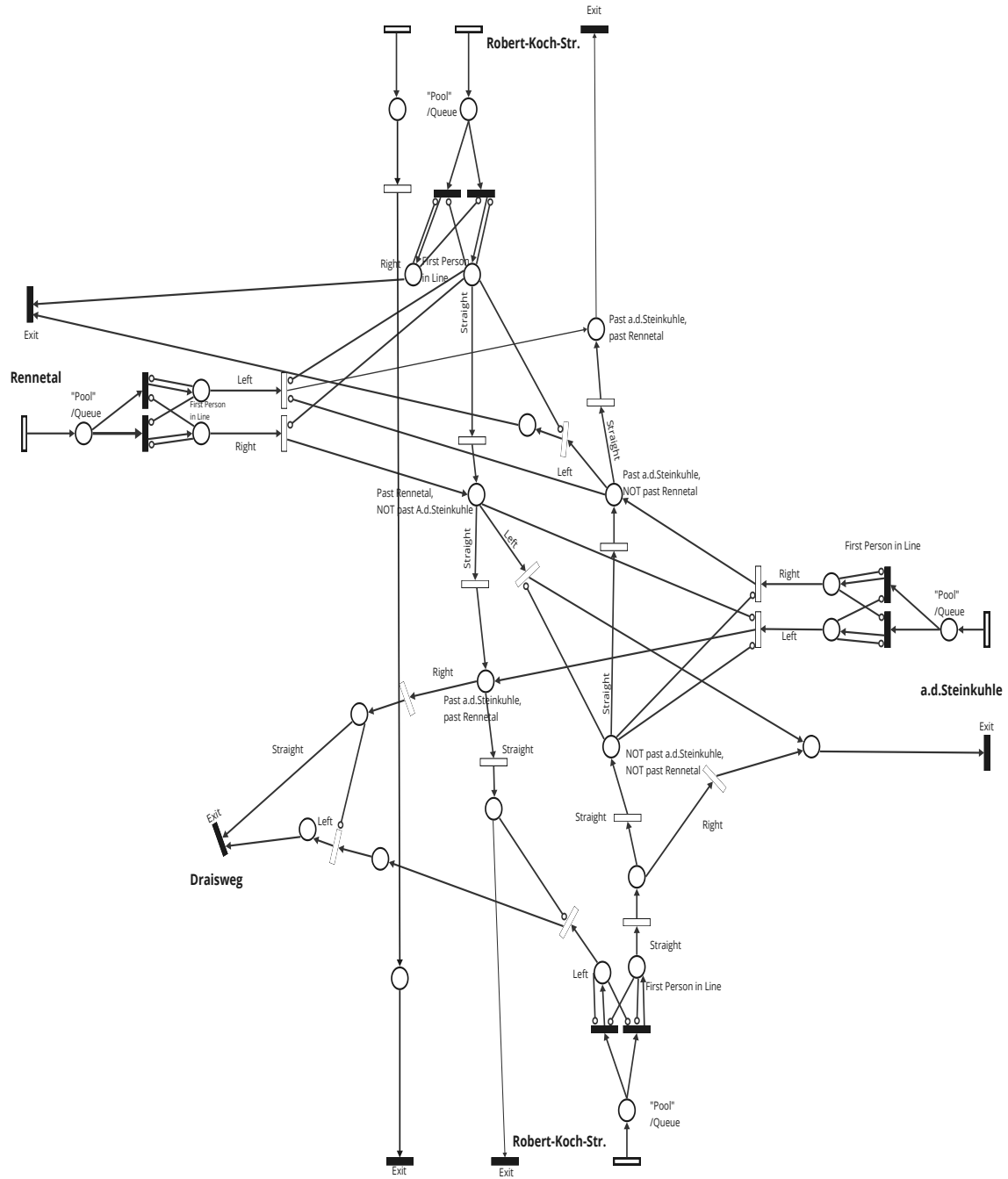


Figure 3.1: Petri net model for the intersection

3.4 Quantities to be measured

We decided on following quantities to be measured:

- **Waiting time** - It is the time for a vehicle in the queue to reach the front.
- **Inter-arrival time** - The inter arrival time is the amount of time between each arrival and the subsequent one in the system.
- **Turn probabilities** - This is the probability that a vehicle which started on one street will turn towards another street in our system.
- **Traffic signal times** - The amount of time needed for a signal to transition between two phases.
- **Throughput** - The number of vehicles passed through the system for a certain amount of time. It helps in validating that our distributions generate a similar number of vehicles in the simulated environment.

3.5 Quantities that will be used as simulation results

The following are the quantities that will be used as the simulation results:

- **Average time to exit simulation** - The average time taken by a vehicle, to enter and pass through the system.
- **Throughput** - The number of vehicles that pass through a node and successfully leave the intersection in a certain amount of time.

4

Data Analysis

Data analysis is mostly done in order to feed information into a simulation model and determine if the result corresponds to the system's real-time output. For the simulation model, we measured the data in real system, approximated the data based on a distribution function and used this function in the simulation program. We collected the data for morning and evening rush hours two times (one for data analysis and one for validation) and measured the inter-arrival times, throughput for cars, bicycles, buses, pedestrians and heavy vehicles in a 15 minute time interval. After collecting the data, we decided its distribution type by plotting histograms (inter-arrival time bins in minutes vs number of cars) and quantile-quantile plots (inter-arrival time in minutes vs F-inverse) for each road. For plotting the histogram, we divided the samples in intervals, plotted the items in each interval and compared the shape of the histograms with known distributions. This was tested by plotting Q-Q plots. Finally, we performed the chi-square test on the guessed distribution. All the distributions obtained were exponential. The results of histograms, Q-Q plots, chi-square tests are given further in section 4.3.

4.1 Data received

The data was provided to us by Landeshauptstadt Magdeburg. It consists of inflow and outflow of vehicles for Robert-Koch-Straße, Rennetal, Draisweg and An der Steinkuhle intersection. The data was recorded on 17.09.2015 (Thursday) from 6AM to 7PM. The data is collected in 15 minutes time interval and contains approximately 50 samples per time interval. The data contains information about car units, buses, trucks, bicycles, light vehicles, total vehicles per hour and vehicle counts. No information about the pedestrians is provided. Also, there is no information about trams since the intersection does not have tram lines. A detailed analysis for 2015 data for Robert-Koch-Straße was also provided to us along with the 2030 prognosis where the intersection expects slightly reduced traffic. We also received two samples of accident data with the details mentioning where and how the accidents occurred along with visualization. Due to the limitations mentioned in section 4.5, we have not used the data given by the city.

4.2 Input and Output data

Given below are the input and output data used for the simulation model.

- **Input data** - Inter-arrival time, turn probabilities of the agent, speed of the agent
- **Output data** - Queue length, throughput, average time spent by the agent in our system

4.3 Results of Histograms, Q-Q plots, Chi-square tests

We started with the assumption of lognormal distribution. With the lognormal distribution, the data did not fit on a straight line in the Q-Q plot. Another assumption of exponential distribution was made where the data plots yielded a straight line passing through the origin for the Q-Q plot. Then a goodness of fit test *i.e.* chi-square test was performed which passed with the significance level of 0.9. The following results represent the inter-arrival times for the vehicles entering the intersection. It should be noted that Draisweg is one-way street and there are no car sources originating from it.

The histograms depicting the number of cars on the Robert-Koch-Straße North is as shown in figure 4.1a. Based on the histogram, we have plotted a Q-Q- plot for the same. It can be observed that, the data points yield a straight line passing through the origin. This signifies that the distribution assumption along with the parameters are right. Further, a chi-square test was also performed to measure the goodness of fit. The parameters are as shown in the table 4.1. This test only helps us know that our assumption of exponential distribution is not wrong since the value of χ_0^2 is lesser than the value $\chi_{f,\alpha}^2$. This suggests that our hypothesis of exponential distribution is not wrong.

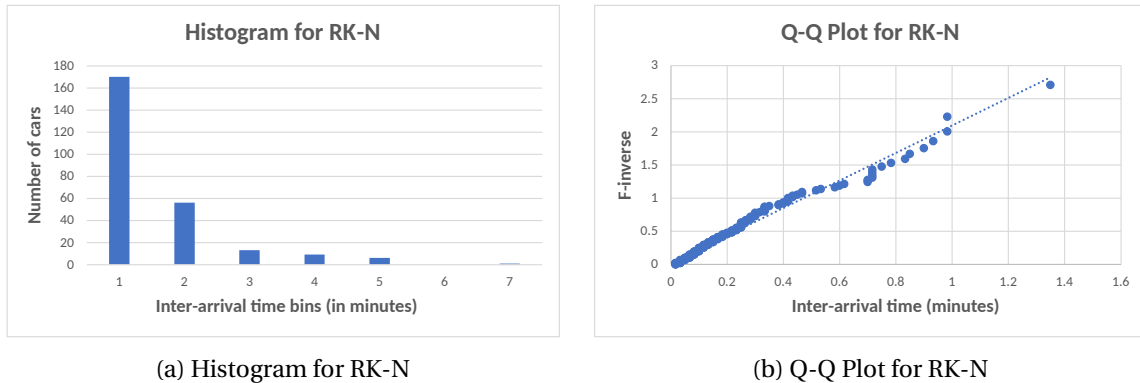


Figure 4.1: RK-N Plots

Parameters	Values
Distribution type	Exponential
Mean(μ)	0.197647059
Lambda(λ)	5.05952381
No of data points(N)	276
Degree of freedom(f)	11
Alpha(α)	0.9
χ_0^2	16.67074986
$\chi_{f,\alpha}^2$	17.27500852

Table 4.1: Chi-square test values for RK-N

Similar to the previous scenario, the histogram (figure 4.2a) was created for cars on Robert-Koch-Straße South and the Q-Q plot (figure 4.2b) was plotted. In this case too, the chi-square test with the parameters as shown in table 4.2 was passed and hence it suggested that our hypothesis of exponential distribution is not wrong.

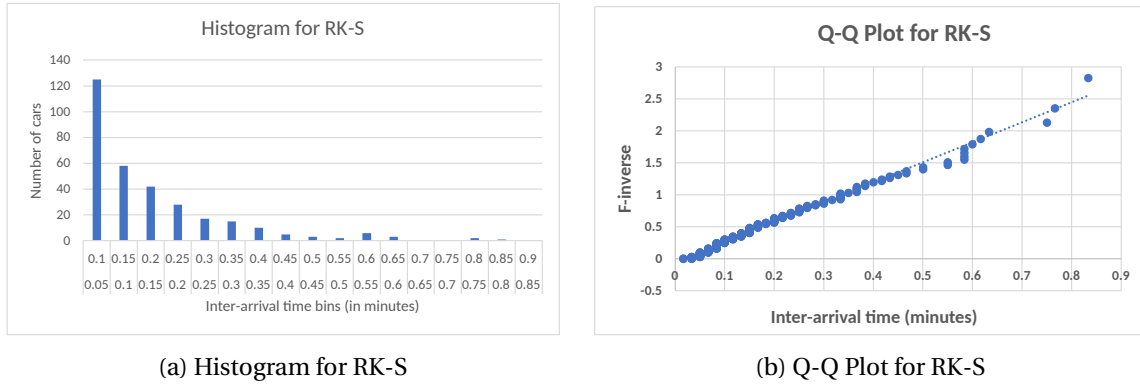
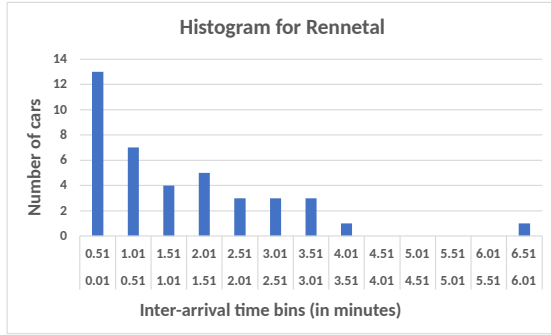


Figure 4.2: RK-S Plots

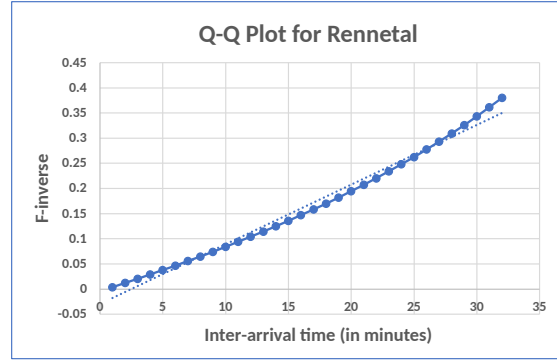
Parameters	Values
Distribution type	Exponential
Mean(μ)	0.157325467
Lambda(λ)	6.35625
No of data points(N)	339
Degree of freedom(f)	14
Alpha(α)	0.9
χ_0^2	18.95431988
$\chi_{f,\alpha}^2$	21.06414421

Table 4.2: Chi-square test values for for RK-S

Similar to the previous scenario, the histogram (figure 4.3a) was created for cars on Rennetal and the Q-Q plot (figure 4.3b) was plotted. In this case too, the chi-square test with the parameters as shown in table 4.3 was passed and hence it suggested that our hypothesis of exponential distribution is not wrong.



(a) Histogram for Rennetal



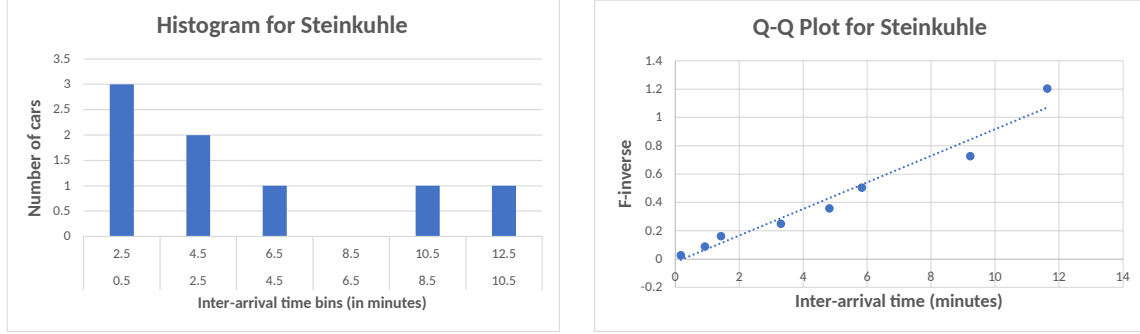
(b) Q-Q Plot for Rennetal

Figure 4.3: Rennetal Plots

Parameters	Values
Distribution type	Exponential
Mean(μ)	1.360833333
Lambda(λ)	0.734843846
No of data points(N)	40
Degree of freedom(f)	10
Alpha(α)	0.9
χ_0^2	10.22805242
$\chi_{f,\alpha}^2$	15.98717917

Table 4.3: Chi-square test values for Rennetal

Similar to the previous scenario, the histogram (figure 4.4a) was created for cars on An der Steinkuhle and the Q-Q plot (figure 4.4b) was plotted. In this case too, the chi-square test with the parameters as shown in table 4.4 was passed and hence it suggested that our hypothesis of exponential distribution is not wrong.



(a) Histogram for An der Steinkuhle

(b) Q-Q Plot for An der Steinkuhle

Figure 4.4: An der Steinkuhle Plots

Parameters	Values
Distribution type	Exponential
Mean(μ)	4.6625
Lambda(λ)	0.214477212
No of data points(N)	8
Degree of freedom(f)	3
Alpha(α)	0.9
χ_0^2	2.714965372
$\chi_{f,\alpha}^2$	6.251388631

Table 4.4: Chi-square test values for An der Steinkuhle

4.4 Difficulties encountered while obtaining the data

It became difficult for us to align all members to collect the data at a particular time. Frequent weather changes and school holidays just before the milestone deadline needed advance planning in data collection. The density of the vehicles started increasing at a particular point which made it difficult for us to keep a track of time. There was a bit disruption in traffic flow as people watched us standing while passing the roads which caused slight deviation. We faced an initial dilemma on deciding what type of data was necessary to build the model. We faced initial difficulty to decide the exact type of distribution needed for few streets. Deciding the common template for entering the collected data in the excel sheet to maintain consistency was difficult at the first time when we initially collected the data.

4.5 Limitations on the accuracy or validity of the data

According to the data provided by Landeshauptstadt Magdeburg, Rennetal road was closed due to construction work on 17.09.2015. The data provided by the city from detailed tables

and overview graphic does not match. The sample size from the city is very small for Rennetal and An der Steinkuhle roads. Since the data is recorded manually by humans, it may have a slight variance.

5

Simulation Program

5.1 Program concept and structure

For writing the simulation program, we used the AnyLogic software. [2] AnyLogic is a simulation modeling software. For the five road intersection, we used the road traffic library for traffic components such as roads, vehicles and buses. We used the process modelling library for logic and entities such as agents which help the model run. The traffic flow of the model originates from 4 car sources and 2 bus sources. The traffic is distributed by 4 select output blocks according to the turn probabilities. The traffic is regulated by 3 intersections and 8 stop lines. The layout of the model was created by using satellite image from Google maps of the intersection by using optimal scaling for drawing roads in AnyLogic. The simulation model is shown in the figure 5.1.

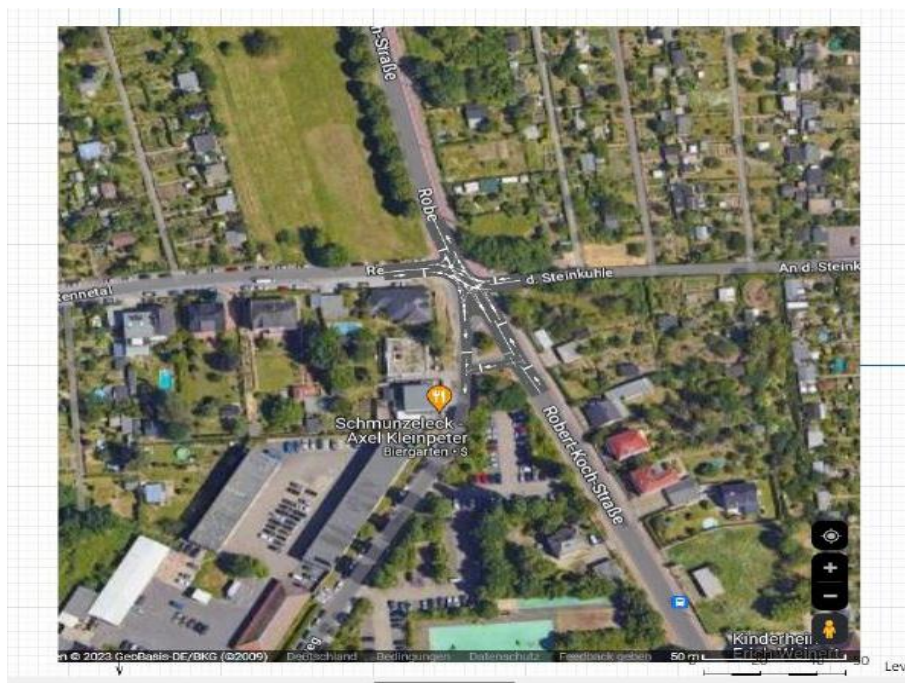


Figure 5.1: Simulation model

5.2 Modularization

The simulation program was designed based on predefined logic and rules which the model and the agent follows to resemble real time scenario. We have the main model, car agent model and the bus agent model. Pedestrian agent is used in experiment 3 which consists of simulating traffic lights.

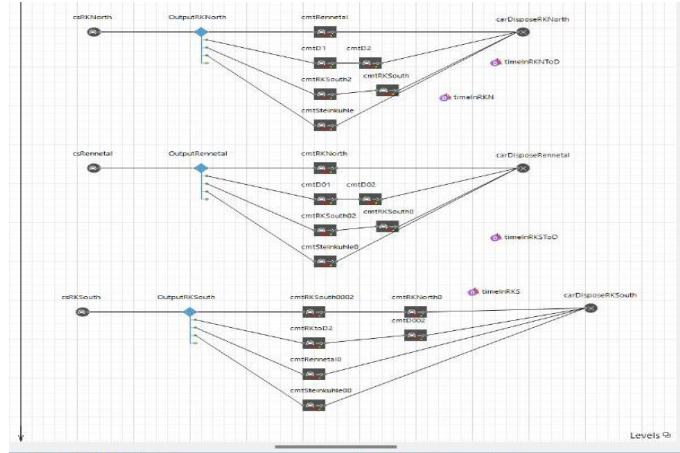


Figure 5.2: Car sources

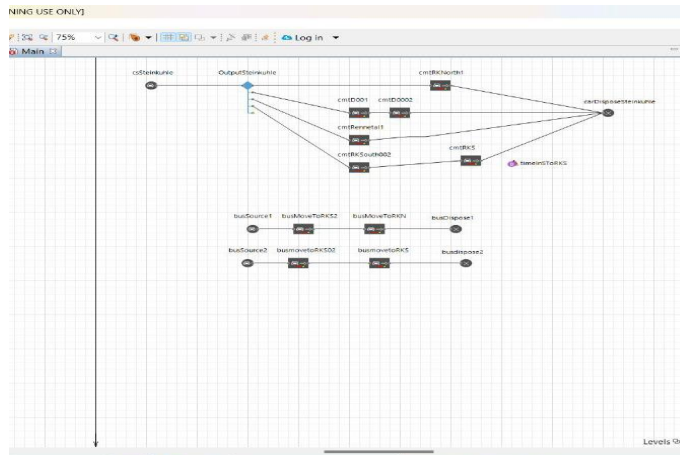


Figure 5.3: Bus sources

5.3 Differences to the conceptual model

There are more transitions for the same road in the conceptual model but in the simulation program, we have one big intersection and stop lines for each road just before the intersection. Speed factor was not taken into account in the conceptual model. Turning probabilities were not taken into account in the conceptual model.

5.4 Steps taken to verify the program

We ran the simulation program with varying parameters to check if the output of the model matches with the real-time collected data. We performed individual block tests while running different agents and modeling different parts of the road. We used common sense for checking if the car sources are modelled correctly. We got our program checked by people who have knowledge about AnyLogic. By use of modular programming, we added different agents and integrated them into the model. By using simple test cases which have known results. For example:- cars does not appear from Draisweg, only 3 buses arrived when running the simulation program for 1 hour. We also performed unit testing where each components were modeled and validated before moving on to the next component.

5.5 Difficulties encountered

Figuring out optimal scaling for drawing the roads in AnyLogic by taking the satellite image from Google Map was difficult. Initial difficulty was faced in using AnyLogic tools. We also faced errors due to incorrect naming of components in the simulation model. It was difficult to model pedestrian crossing as pedestrians move on their own and also cars overtaking and hence we did not model pedestrians in our simulation model.

6

Validation

Validation involves comparing the simulation model with the actual system. It entails building the correct model. The decisions that are made based on the simulation results are the same as if we had been able to use results from the real system instead.

We have followed a two-step validation - developed a model with high face validity and then determined the quality of the simulation output.

The objective of validation was to check the reliability, accuracy, usability and confidence in the simulated model.

6.1 Quantities of interest

Following are our quantities of interest:

- **Throughput** - The number of vehicles that pass through a node and successfully leave the intersection in a specified amount of time.
- **Time spent by the vehicles** - Time spent by the cars on a specified road to enter and exit the system.

6.2 Sensitivity Analysis

Face validity means the results appear reasonable to model users and experts. One of the popular approaches, which we have used is the **Sensitivity Analysis**.

Formula:

$$\begin{aligned} & \text{Outputvalue (O)} \\ & \text{Inputparameter (I)} \\ & \text{Sensitivity (S)} = \frac{dO}{dI} \end{aligned} \tag{6.1}$$

In this, we are evaluating the extent to which the output values depend on the input parameters. In our simulation model, we have added a parameter named “traffic volume”, which influences the inter-arrival times for all of the streets in the different ways:

- **Traffic volume** 0.5 implies multiplying the inter-arrival times with 2 which infers to half the amount of cars.
- **Traffic volume** 1 implies regular distributions from data analysis.
- **Traffic volume** 2 implies multiplying the inter-arrival times with 0.5 which results in double the amount of cars.

The "Parameter Variation Experiment" feature in AnyLogic allows us to choose a range for the traffic volume and have the experiments run repeatedly with different input parameters.

6.3 Model parameters for sensitivity analysis

We have used the following model parameters for sensitivity analysis:

- Model run time for each replication: 15 mins
- Number of replications: 100
- Values for traffic volume: [0.5, 2] with step size: 0.1 $\rightarrow$ 16 Iterations

6.4 Sensitivity analysis results

There were not many vehicles snarled in a traffic bottleneck because doubling the number of cars entering the intersection results in practically doubling the system's throughput. This can be observed in the figure 6.1.

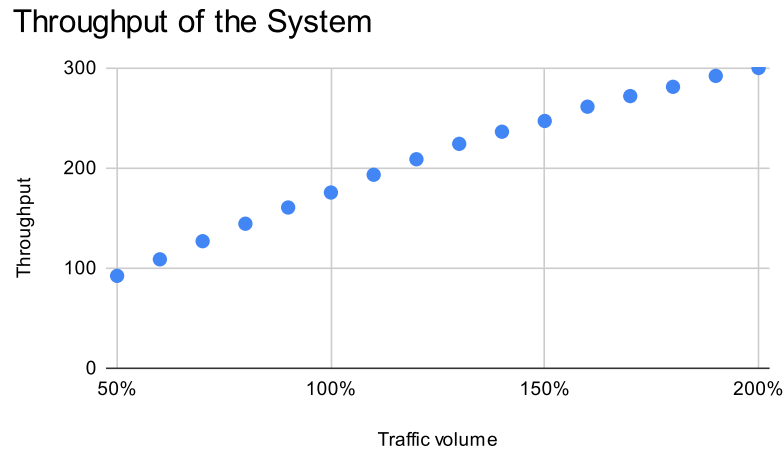


Figure 6.1: Throughput of the system on varying traffic volume

Our system is not very sensitive to a change in the input variables, as seen by the fact that the mean time only marginally increases for an increase in traffic volume. This can be in the figure 6.2 below.

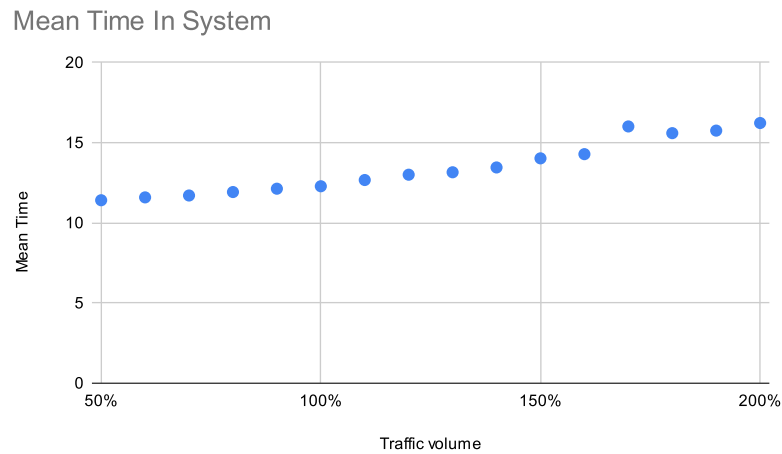
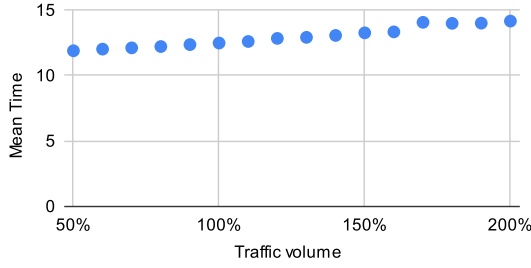


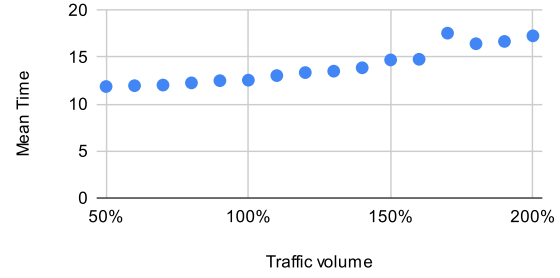
Figure 6.2: Mean time of vehicles in the system

Because we did not actually notice queues when we collected the data in the real-world system, and the majority of the traffic could easily flow through it, the results below in figure 6.3 appear reasonable. The intersection shouldn't be overcrowded and waiting times shouldn't grow even with a doubling of the number of vehicles.

Robert-Koch-Str. North to South



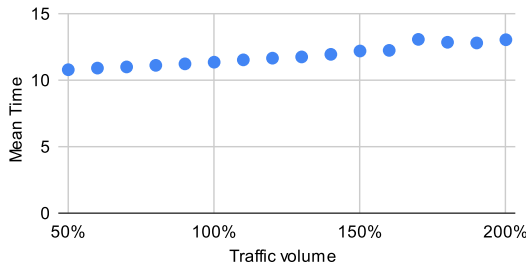
Robert-Koch-Str. South to North



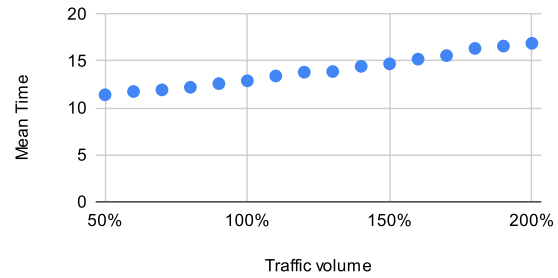
(a) Mean time for vehicles from Robert-Koch-Straße North to South

(b) Mean time for vehicles from Robert-Koch-Straße South to North

RKN to D



RKS to D



(c) Mean time for vehicles from Robert-Koch-Straße North to Draisweg

(d) Mean time for vehicles from Robert-Koch-Straße South to Draisweg

Figure 6.3: Mean times for vehicles of the streets

6.5 Input-Output transformations and results

Input-Output Transformations are used to compare the results from the simulation model to the results from our real system. In this validation approach, we use idealised input data, in other words, random variables. The simulation model ran for 25 replications with a run time of 15 mins for each replication. We started the analysis taking confidence level 90%. We further checked with confidence level 98% and found it showed a reasonable result. Hence, taking the confidence level as 98% and level of significance (α) as 0.02, we found the upper and lower bound of confidence interval for each output variable. Then we checked the position of real world mean of the selected output variables in the confidence interval created to see if it lies inside the interval. These can be seen in the figure 6.5 and figure 6.4 below, where we have a boxplot of the time spent in the system comparing the simulated results with the real world system.

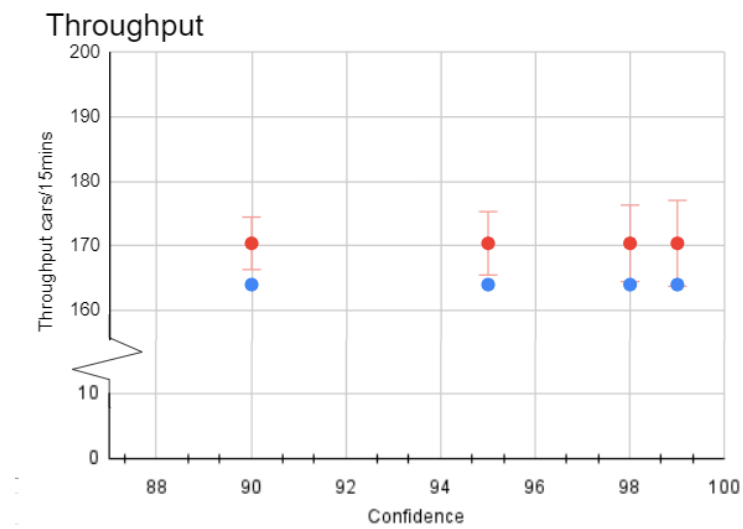


Figure 6.4: Boxplot of the throughput of the systems

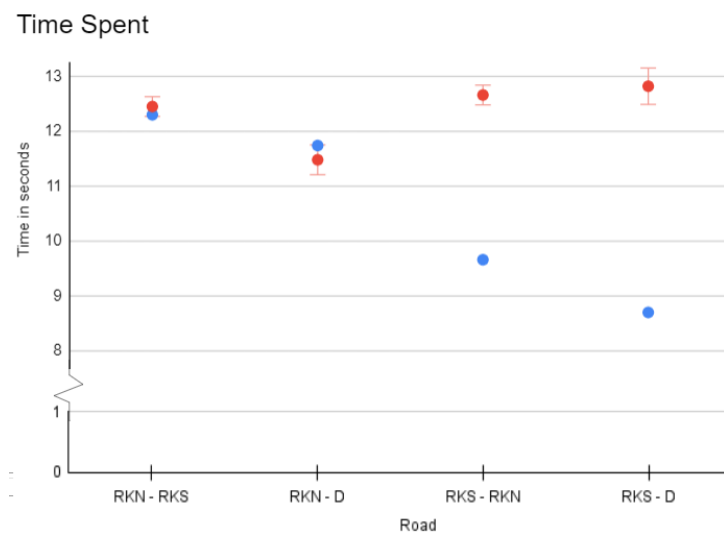


Figure 6.5: Boxplot of the time spent in the systems

6.6 Justifications

Throughput

The above plot for throughput clearly depicts that with confidence level of 98% and alpha 0.02, the real world mean lies inside the confidence interval created from the output variable for the same. It adds value to our attempt to show the correctness of the simulation model.

Time spent

From the above figures 6.4 and 6.5, it is evident that the real world mean of time spent by cars on the Road RKN - RKS and RKN - D lies clearly inside the confidence interval. But for the roads RKS - RKN and RKS - D, real world mean lies outside the confidence interval. Upon analysing the model, we found that the cars in simulation model on these roads slows down more when they reach the intersection which is not observed in the real world. This is the reason why the real world mean lies outside the interval with a margin of 3 - 5 seconds. Considering safety into account, we expect the cars to slow down when they reach intersection as in the simulation model and thereby we decided to remove the output variables for time spent on the roads RKS - RKN and RKS - D from scope of validation.

6.7 Statement of confidence

The validation experiment is done with 25 replications, 98% confidence level and the significance level of 0.02, which supports the accuracy and the reliability of the model. Sensitivity analysis yielded positive results, demonstrating its reliability and robustness across a wide range of parameter values.

6.8 Changes made to the model

In order to correlate our measuring positions while gathering the data, we decreased the length of the streets in our model. In accordance with our distributions, we also had to modify our model's parameter variation for the inter-arrival times ("trafficVolume"). There was fixed scaling in our model in comparison to the map of the intersection. The street speed limit was used to modify vehicle speeds. Our model's stop lines were also adjusted to get the right results.

6.9 Limitations to the scope of validity

There is a limited amount of data to form a queue in the model and also no traffic lights, due to which it is really difficult to validate the queue length. As the intersection is small, the

waiting time for vehicles is less. Since there is no zebra crossing, bicycles and pedestrians are free to cross the street whenever they feel secure to do so.

Simulation Experiments

This section contains a detailed explanation on the series of experiments conducted on the baseline simulation model. The experiments were conducted with an aim to improve the efficiency of the current system.

7.1 Experiment 1 - One Large Roundabout

7.1.1 Idea

The first experiment is creating one big roundabout combining all streets of the intersection namely Robert-Koch-Straße, Rennetal, An der Steinkuhle, and Draisweg. The goal is to implement continuous traffic flow and reduce waiting time/conflicts at the intersection. This experiment was intended on making the intersection safer for motorists.

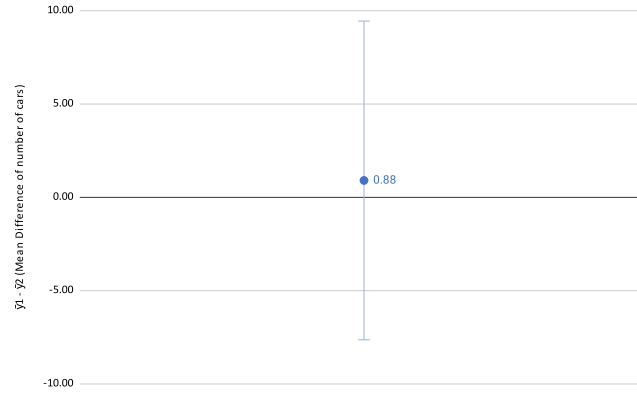


Figure 7.1: One roundabout combining all streets

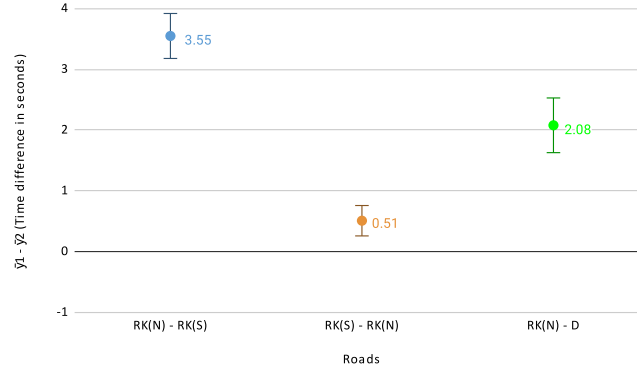
7.1.2 Results obtained after keeping the same traffic volume

By keeping the traffic volume same, we performed correlated sampling to compare the performance of validated simulation model and experimental model. The number of replications performed are 25. The level of significance is 0.02. With 98% confidence level we calculated the confidence across the difference in mean values ($\bar{y}_1 - \bar{y}_2$). Here $\bar{y}_1$ is the mean of experiment 1 values and $\bar{y}_2$ is the mean of validated model values. If the interval lies in the positive quadrant for throughput, this means that the experiment model performs better than the validated model. From the figure 7.2a, we can see that 0 lies in between the confidence interval which suggests that the result for throughput is not statistically significant.

From figure 7.2b, it is clear that the time spent by the cars in the experiment are more than the validated model for all roads. Since all the three confidence intervals do not include zero, all the results are considered statistically significant.



(a) CI for mean difference in throughput



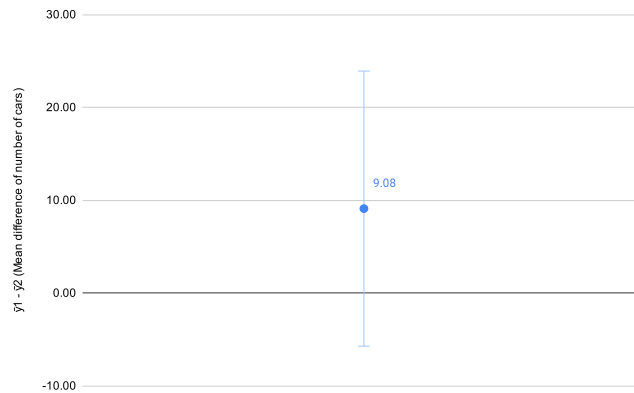
(b) CI for mean difference in time spent

Figure 7.2: Confidence interval for experiment 1

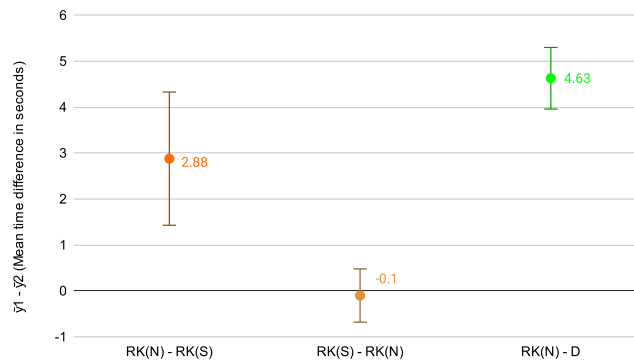
7.1.3 Results obtained after doubling the traffic volume

As we can see from the figure 7.3a, increase in throughput is not statistically significant as the interval contains zero.

The time spent for roads RK(N)-RK(S) and RK(N)-D in the figure 7.3b are statistically significant but the time spent by the cars in the experiment model is more than the validated model which is not the expected outcome.



(a) CI for mean difference in throughput



(b) CI for mean difference in time spent

Figure 7.3: CI for Experiment 1 with doubled traffic volume

7.1.4 Recommendations

Based on the experiment results, there was a slight increase in time spent by the cars in the system which was not the desired expected output. It had no other positive significant impact on the overall system. We doubled the throughput of the system to check if the proposed roundabout performs better than the intersection. But as explained in the above section, the performance was not significantly better. The cost of planning and

construction of the roundabout will not yield a good return on investment. Hence this experiment is **not recommended**.

7.2 Experiment 2 - Two Small Roundabouts

7.2.1 Idea

This experiment involves two smaller roundabouts, one combining R-K-Straße and Draisweg and another combining R-K-Straße and An der Steinkuhle, Rennetal. The goal was to implement continuous traffic flow, reduce waiting time and conflicts at the intersection. This was designed to be an alternative for experiment 1 (One big roundabout). The idea was to investigate the effects of smaller roundabouts on the system by replacing a very large roundabout and also verify if smaller intersections work more efficiently than a big one. This need was felt when it was observed that the cars going from R-K-Straße North to An der Steinkuhle and cars going from R-K-Straße South to Rennetal had to travel all the way around the big intersection.

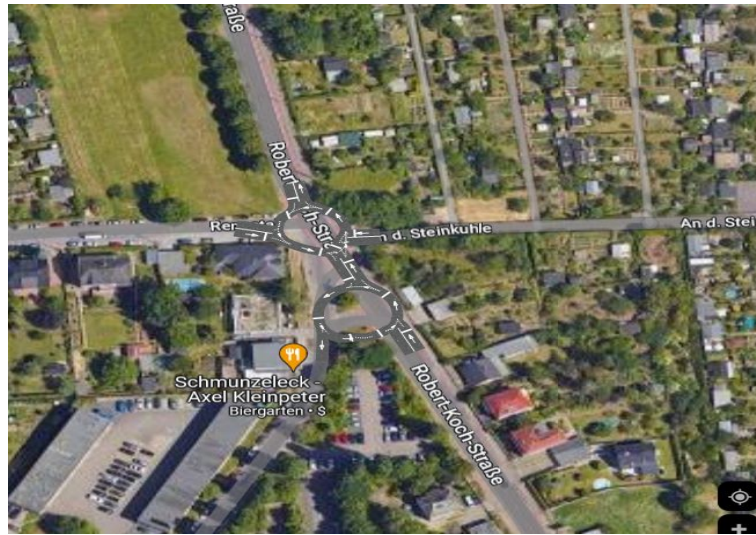


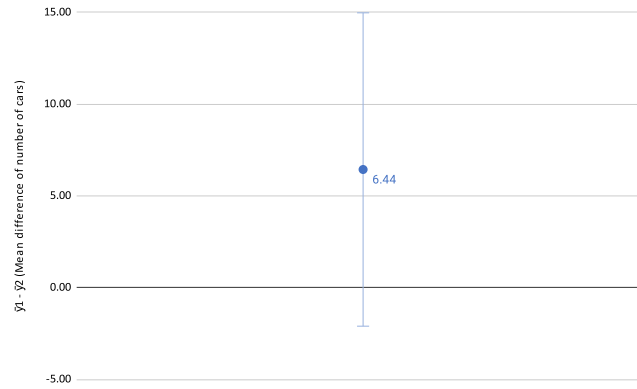
Figure 7.4: Two roundabouts

7.2.2 Results obtained after keeping the same traffic volume

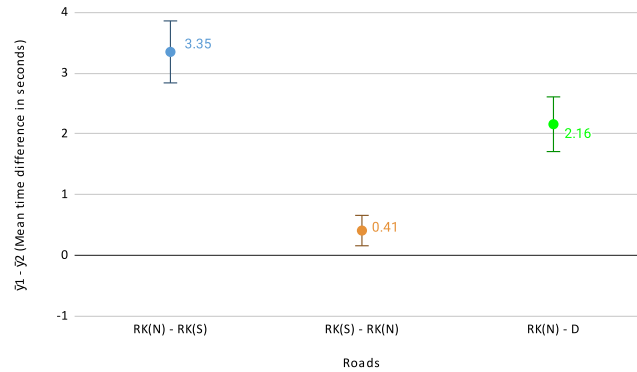
By keeping the same traffic volume, we performed correlated sampling to compare the performance of validated simulation model and experimental model. The number of replications performed are 25. The level of significance is 0.02. With 98% confidence level we calculated the confidence across the difference in mean values ($\bar{y}_1 - \bar{y}_2$). Here $\bar{y}_1$ is the mean of experiment 2 values and $\bar{y}_2$ is the mean of validated model values. If the interval lies in the positive quadrant for throughput, this means that the experiment model

performs better than the validated model. From the figure 7.5a, we can see that 0 lies in between the confidence interval which suggests that the result for throughput is not statistically significant.

From figure 7.5b, it is clear that the time spent by the cars in the experiment are more than the validated model for all roads. Since all the three confidence intervals do not include zero, all the results are considered statistically significant. This output suggests that the time spent by the cars in the roundabout is increased but does not have any significant increase in throughput.



(a) CI for mean difference in throughput



(b) CI for mean difference in time spent

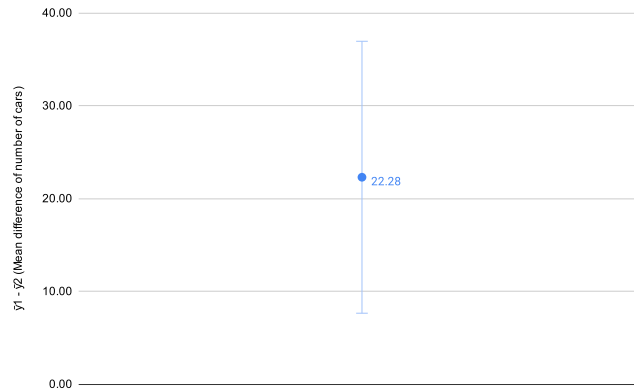
Figure 7.5: Confidence interval for experiment 2

7.2.3 Results obtained after doubling the traffic volume

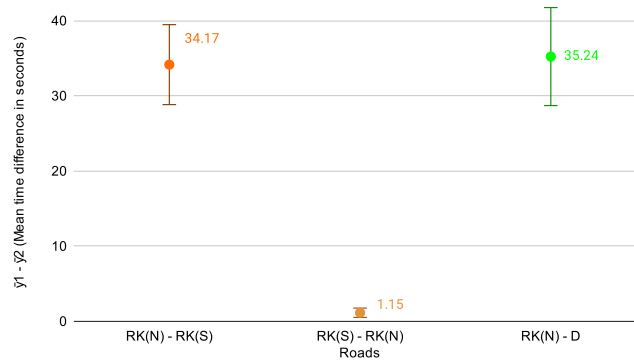
As we can see from the figure 7.6a, increase in throughput is statistically significant as the interval does not contains zero. The throughput is increased by 22 cars which is a positive indicator that the experiment is performing better than the validated model in the scenario where traffic volume is doubled.

The time spent for roads RK(N)-RK(S) and RK(N)-D in the figure 7.6b are statistically

significant but the time spent by the cars in the experiment model is more than the validated model which is not the expected outcome.



(a) CI for mean difference in throughput



(b) CI for mean difference in time spent

Figure 7.6: CI for Experiment 2 with doubled traffic volume

7.2.4 Recommendations

This experiment did not yield better results than experiment 1 and validated model. The observed outputs were similar to the outputs of experiment 1 suggesting that two roundabouts is not an improved alternative to a large single roundabout. In the scenario where we doubled the traffic volume as discussed above, the throughput of the system increases significantly. But this positive improvement has a negative effect on the time spent by the cars. Therefore, the improvement in throughput is not valuable when the cars have to spend for more time in the system. Moreover, the cost of planning and construction of two roundabouts would not yield a good return on investment. Hence, this experiment is **not recommended**.

7.3 Experiment 3 - Two Traffic Lights

7.3.1 Idea

The idea is to have two traffic lights on the north and south side of Robert-Koch-Straße. We used the following traffic signal settings - 30 seconds for the green light, 7 seconds for yellow light and 30 seconds for red light. This aims at providing a safe environment for the pedestrians to cross the heavily active Robert-Koch-Straße.



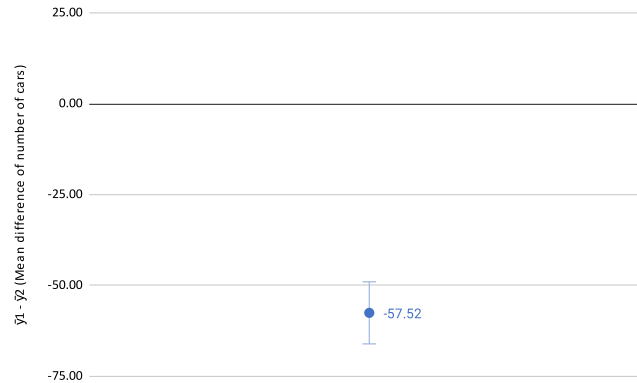
Figure 7.7: Two traffic lights on Robert-Koch-Straße

7.3.2 Results and Observations

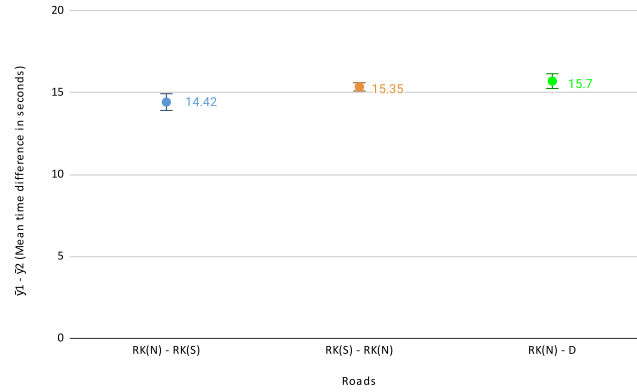
This experiment provides a safe environment for the pedestrians to cross the streets. But there is more than 100% increase in the time spent by the cars. The throughput of the system is also reduced by 30%. We performed correlated sampling to compare the performance of validated simulation model and experimental model. The number of replications performed are 25. The level of significance is 0.02. With 98% confidence level we calculated the confidence across the difference in mean values ($\bar{y}_1 - \bar{y}_2$). Here $\bar{y}_1$ is the mean of experiment 3 values and $\bar{y}_2$ is the mean of validated model values. If the interval lies in the negative quadrant for throughput, this means that the experimental model performs worse than the validated model. From the figure 7.8a, we can see that 0 does not lie inside the confidence interval which suggests that the result for throughput is statistically significant.

From figure 7.8b, it is clear that the time spent by the cars in the experiment are more than the validated model for all roads. Since all the three confidence intervals do not include zero, all the results are considered statistically significant. This output suggests that the

time spent by the cars in the intersection is increased and the throughput decreases drastically too.



(a) CI for mean difference in throughput



(b) CI for mean difference in time spent

Figure 7.8: CI for Experiment 3

7.3.3 Recommendations

Implementing a pedestrian crossing does not justify almost doubling the wait time for motor vehicle traffic. This also reduces the throughput of the intersection drastically. Hence, this experiment is **not recommended**.

7.4 Experiment 4 - Removing One-Way Restriction

7.4.1 Idea

The goal of this experiment is to remove the one-way restriction on Draisweg and to ease the traffic volume on the main road[RK(S) to RK(N)].

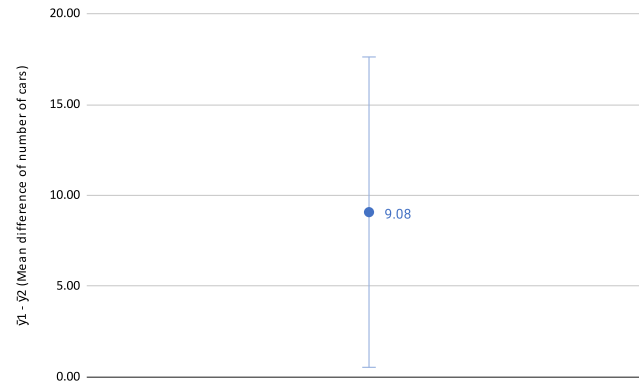


Figure 7.9: Removing one way restriction

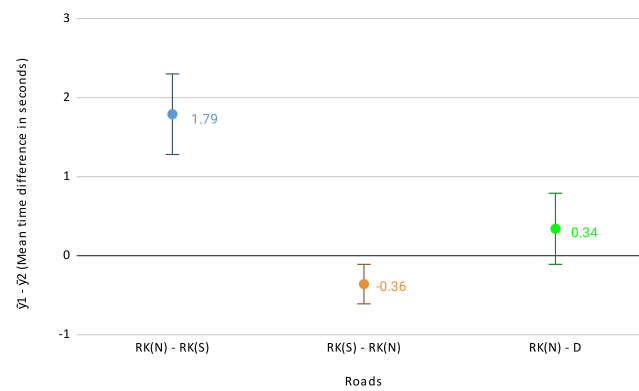
7.4.2 Results and Observations

This experiment has increased the throughput of the overall system. But there is no significant rise in the time spent by cars. It will also complicate the intersection for vehicles and pedestrians. We performed correlated sampling to compare the performance of validated simulation model and experimental model. The number of replications performed are 25. The level of significance is 0.02. With 98% confidence level we calculated the confidence across the difference in mean values ($\bar{y}_1 - \bar{y}_2$). Here $\bar{y}_1$ is the mean of experiment 4 values and $\bar{y}_2$ is the mean of validated model values. If the interval lies in the positive quadrant for throughput, this means that the experimental model performs better than the validated model. From the figure 7.10a, we can see that 0 does not lie inside the confidence interval which suggests that the result for throughput is statistically significant.

From figure 7.10b, it is clear that the time spent by the cars at RK(N)-RK(S) in the experiment are more than the validated model. The result for the road RK(N)-D is not statistically significant because it includes zero in the interval. There is a slight improvement in performance for RK(S)-RK(N). This output suggests that the condition improves only for the road RK(S)-RK(N) and has negative effects on the rest of the roads.



(a) CI for mean difference in throughput



(b) CI for mean difference in time spent

Figure 7.10: CI for Experiment 4

7.4.3 Recommendations

This experiment will unnecessarily complicate the intersection with no added benefits. This will also make the neighbourhood on the Draisweg road unnecessarily more noisy. Hence, this experiment is **not recommended**.

7.5 Recommendation Summary

Even after doubling the traffic volume our experiments performs similar to the current intersection. After further output analysis and comparison, we found out that the current existing intersection can manage even if there is gradual increase in the traffic volume over the years. The experiments with the roundabouts were initially promising, but upon doubling the traffic volume for further testing it was observed that there was a significant compromise in the time spent and the overall throughput of the vehicles in the model.

None of the experiments provide a good enough return on investment. Therefore, we recommend not to make any changes to the intersection.

8

Challenges Encountered

8.1 Challenges encountered

There were several challenges encountered during the execution of the project, which are listed as follows:

- With everyone's schedule, it was challenging to plan and arrange.
- It was difficult to figure the right way to start the project.
- It was challenging to translate and comprehend the information provided by the city because of the presence of technical terms.
- It was laborious and time-consuming to decide what data needed to be gathered.
- It was difficult to come up with strategies to validate the model.
- The inability to move on to the next milestone was caused by the backlog of a few milestones.
- Human errors are inevitable while collecting data.
- Getting the data to fit the distributions proved difficult.

Project Cost Diagram

The figure 9.1 shows the planned and actual cost incurred for each milestone. As seen in the figure, the planned and actual cost required for milestones 1 and 8 were almost the same. However, for data analysis, project plan and experiments the actual cost went up than the planned cost and vice versa for conceptual model, simulation model and validation. We have finished the project effectively within the allocated budget. We also received 10 hours for best presentation in Milestone 2. Hence our updated budget was 61.000€. Our total cost came up to 55.200€, which is well under the total budget. It is also shown in the figure 9.1.

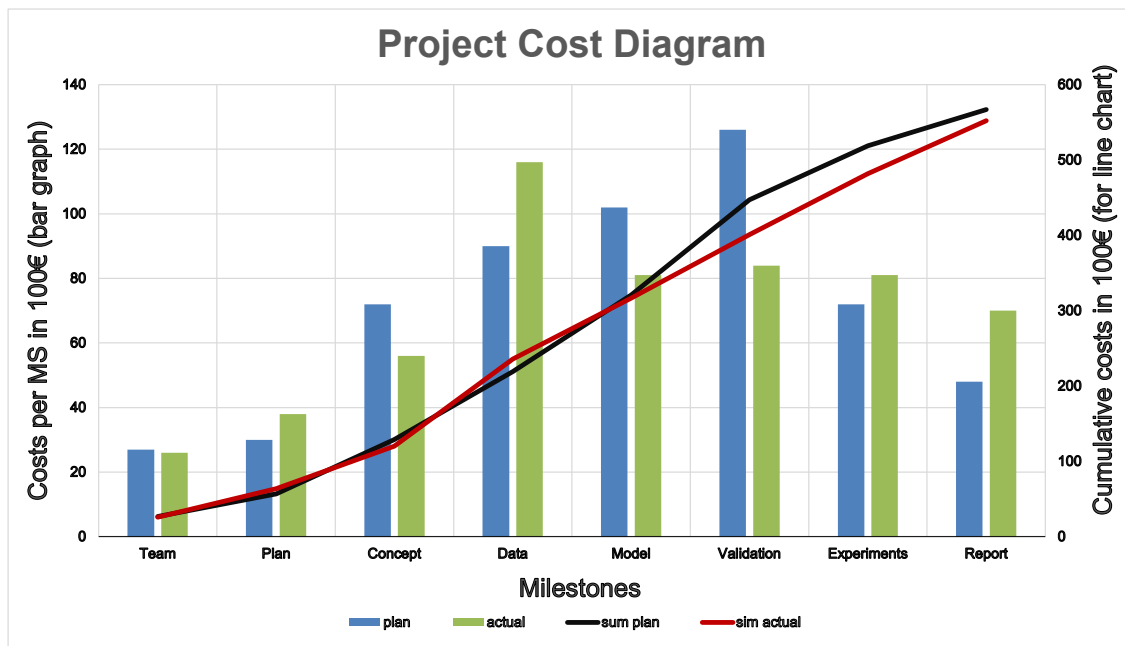


Figure 9.1: Project cost diagram

10

Project Progress Diagram

We kept a track of our project progress through the project progress diagram as seen in the figure 10.1. The milestones created were a very creative way to keep our progress in check. We were on track until milestone 3. However, we lagged in milestone 4, due to public holidays and lack of proper planning to collect the data. But, we were again back on track from milestone 5 and were ahead of the planned progress and have completed the project as expected in time.

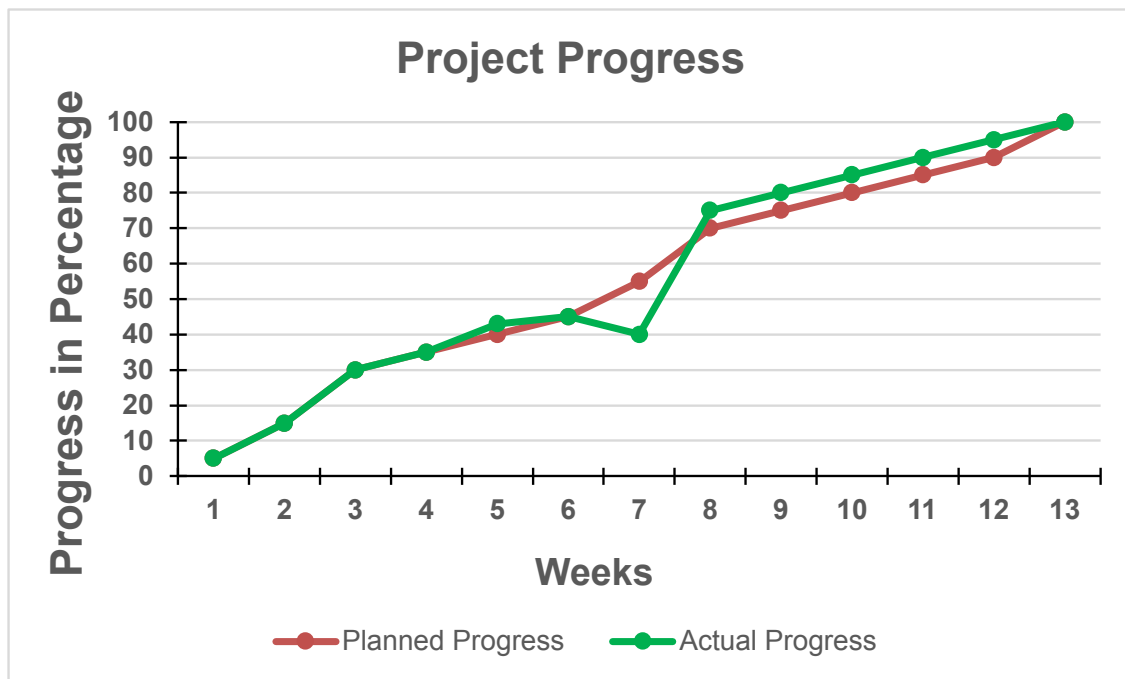


Figure 10.1: Project progress diagram

11

Conclusion

In conclusion, we utilized AnyLogic software to simulate and test various alterations to the Robert-Koch-Straße, Rennetal, An der Steinkuhle and Draisweg intersection to improve its efficiency. By varying the signal times, throughput, turning probabilities and lane configurations interesting trends were observed. The results from the experiments highlight the importance of stress testing the real and modified models before investing capital to make physical changes to the intersection. An important assumption that we made is that the driver behavior remains consistent and follows all traffic rules. This may differ from real-world scenarios. After analyzing the results for all the 4 experiments, we focused on 2 promising ones, which are the intersection with the large roundabout change and the intersection after introduction of the traffic lights. However, after further output analysis and comparison, it was clearly observed that the systems could handle higher traffic volumes especially the ones with the roundabouts but there was a significant compromise in the time spent and the overall throughput of the vehicles in the model. Based on these results we feel that the cost of implementing the changes in the real world is not worth due to the slight increase in efficiency it produces. Moreover, based on the city data for accidents, it reports that there are very few. It also tells us that the city expects that the traffic volume in the intersection is going to remain the same over the next decade. So, we conclude that the intersection would be better off without changes proposed in the experiments.



Bibliography

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